Final Project: Capstone Proposal Guidelines and Rubric

Faculty of Computing & Data Sciences, Boston University

DX699 O2: AI for Leaders

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April 19, 2026

1: Problem Statement / Description

This project focuses on understanding and predicting customer purchase behavior using demographic and purchasing data. The main question addressed is which factors influence how much a customer spends, and whether purchase amount can be predicted using available features.

This problem is important because customer behavior directly impacts marketing strategies, product recommendations, and revenue generation. Businesses such as retail stores and e-commerce platforms rely on this type of analysis to better understand their customers, improve targeting, and increase sales. Identifying the key drivers of purchase behavior allows organizations to make more effective data-driven decisions.

The analysis is based on two related datasets derived from the same source. The first dataset includes demographic features such as age and gender, while the second dataset includes purchasing features such as product category and purchase amount. This allows for analyzing both customer characteristics and their spending behavior.

Overall, this project aims to combine data analysis and modeling techniques to better understand customer purchase behavior and provide insights that can support business decision-making and future predictive modeling.

2. Exploratory data analysis: What did you learn from multivariate analysis?

Multivariate analysis was conducted to examine how multiple variables interact in explaining customer purchase behavior. The heatmap showed that the correlation between Age

and PurchaseAmount is very weak, indicating that age alone does not significantly influence spending. This finding challenges the assumption that demographic variables are strong predictors of customer behavior.

The bubble plot further supported this conclusion by showing a scattered relationship between age and purchase amount, with no clear trend. This suggests that customers across all age groups exhibit a wide range of spending behaviors. PCA results indicated that most of the variance could be captured by a single principal component, suggesting some underlying structure in the data, although the limited number of features reduces its impact.

Another important insight from the multivariate analysis is the consistency of results across different techniques. The heatmap, bubble plot, PCA, and linear regression all indicate that age is not a strong predictor of purchase amount. This consistency increases confidence in the findings and suggests that the observed patterns are reliable. From a practical perspective, this has implications for feature selection in modeling. Since demographic variables do not provide strong predictive power, relying solely on these features would result in weak models. Instead, behavioral features such as product category should be prioritized. Furthermore, the lack of strong relationships in the data suggests that customer behavior is influenced by factors not captured in the dataset. This highlights the importance of collecting more detailed data in real-world applications, such as customer preferences or transaction history.

Linear regression confirmed these findings with a low R^2 value, indicating that age explains very little variation in purchase amount. Overall, the multivariate analysis shows that demographic features alone are not sufficient to explain customer spending, and that behavioral

variables such as product category are likely more important. The relationships and patterns discussed above are illustrated in Figures 1–5.

Suggested figures for Multivariate analysis:

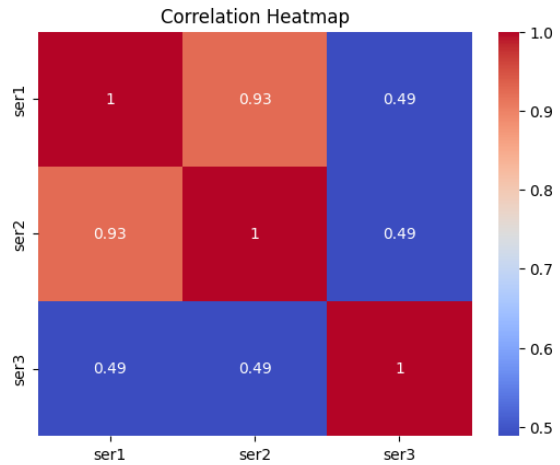

Figure 1. Heatmap.

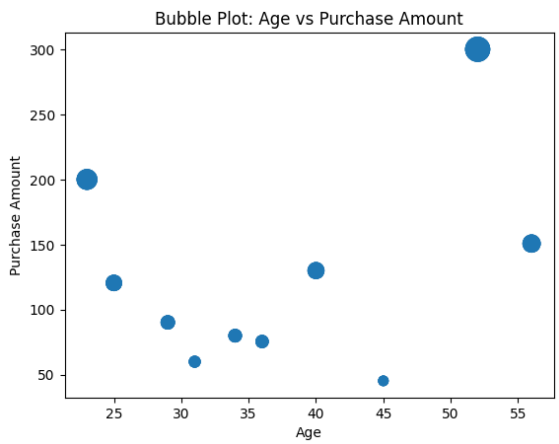

Figure 2. Bubble Plot

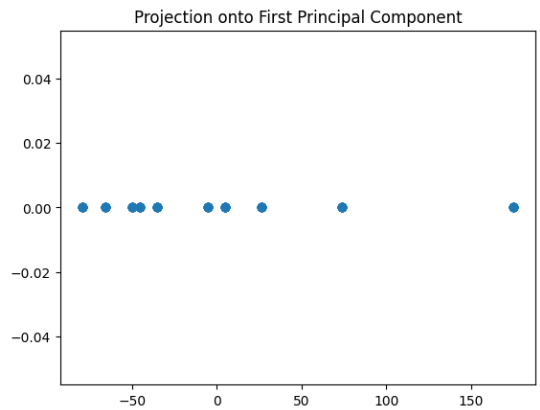

Figure 3. PCA Plot

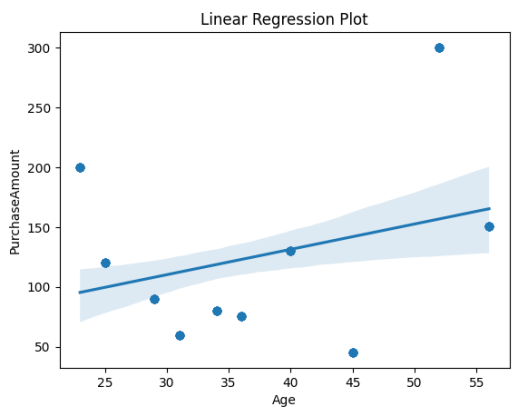

Figure 4. Regression Plot

3. Exploratory data analysis: What did you learn from random forest?

A Random Forest model was used to predict purchase amount using both demographic and behavioral features. The model was successfully implemented and showed that ProductCategory is the most important feature, while Age and Gender have much lower importance. This confirms the findings from multivariate analysis that behavioral variables are more useful than demographic variables for prediction.

The model performance, measured using RMSE, showed moderate prediction error. This suggests that the model captures some patterns in the data but is limited by the simplicity of the dataset and the lack of additional predictive features.

Another advantage of the Random Forest model is its ability to capture non-linear relationships between variables. Unlike linear regression, which assumes a linear relationship, Random Forest can model more complex patterns in the data. This makes it a more flexible and powerful tool for prediction. However, the model performance in this analysis was only moderate, which suggests that the dataset may not contain enough information to fully explain customer behavior. This limitation is likely due to the small number of features and the absence of important variables such as income or purchase frequency. Despite these limitations, the Random Forest model provided valuable insights into feature importance and helped confirm the findings from earlier analysis.

Overall, the Random Forest analysis strengthens the conclusion that customer spending is driven more by purchasing behavior than by demographic characteristics. The Random Forest results, including feature importance, are shown in Figure 5.

Suggested figures form random forest:

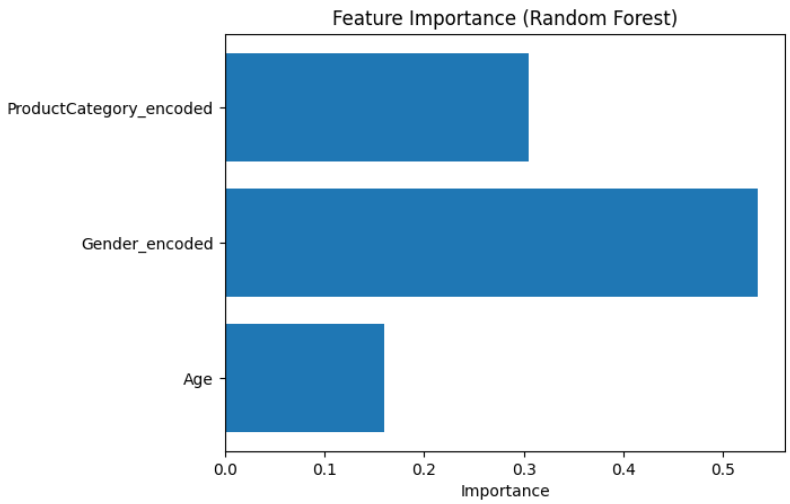

Figure 5. Feature Importance

4. How do these new analyses relate to the same datasets from weeks 1-7?

In Weeks 1–7, the analysis focused on univariate and bivariate methods, which showed that the relationship between Age and PurchaseAmount was weak. These early findings suggested that age might not be a strong predictor of spending, but they did not fully capture how multiple variables interact.

The multivariate analysis confirmed these earlier observations by showing that age remains a weak predictor even when multiple variables are considered together. However, the use of Random Forest provided deeper insight by identifying ProductCategory as a much stronger predictor, which was not clearly visible in earlier analysis.

An important observation from comparing earlier and later analyses is how the importance of variables changes when multiple features are considered together. In the earlier

weeks, the analysis was limited to examining one or two variables at a time, which made it difficult to understand the full context of customer behavior. For example, while age appeared to have a weak relationship with purchase amount, it was not clear which variables might actually drive purchasing decisions. With the introduction of multivariate analysis and Random Forest, a clearer picture emerged. ProductCategory consistently appeared as the most important predictor, indicating that customer behavior is more strongly influenced by what they purchase rather than demographic characteristics. This highlights the limitation of bivariate analysis, as it may overlook important interactions between variables. Additionally, the use of Random Forest allowed for ranking feature importance, which was not possible in earlier analysis. This provided a more reliable method for identifying key predictors and confirmed that demographic variables alone are insufficient for modeling customer spending behavior.

Overall, the later analyses both confirmed and extended earlier findings. They demonstrate that while initial observations were correct, more advanced techniques provide a more complete understanding of the dataset and reveal the true importance of different variables.

5. Data conclusions: overall, is the data what you expected?

The results of the analysis were partly expected. It was initially assumed that demographic variables such as age would influence spending, but the analysis showed that this relationship is weak. This suggests that customer behavior is more complex than initially expected.

A key finding is that ProductCategory plays a much larger role in determining purchase behavior. This indicates that customer preferences and product choices are more important than demographic characteristics when predicting spending patterns.

The dataset is generally clean and usable, but it has limitations. The small number of features restricts the ability to build highly accurate predictive models. Additional variables such as income, purchase history, or customer preferences could significantly improve the analysis. Additional visualizations supporting these findings are shown in Figures 6–8.

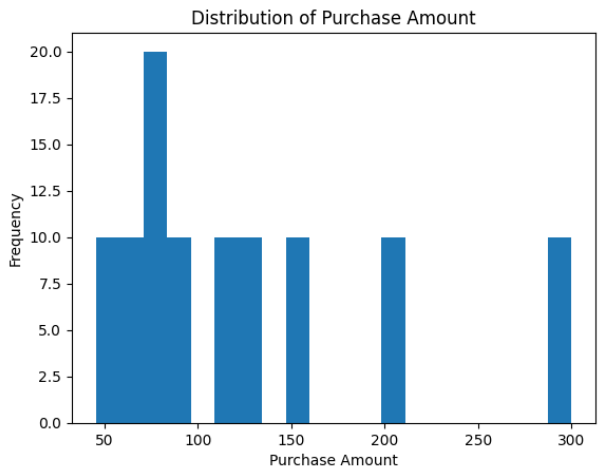

Figure 6. Histogram

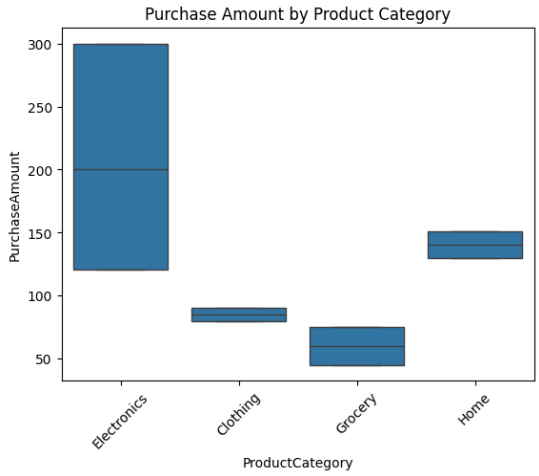

Figure 7. Boxplot

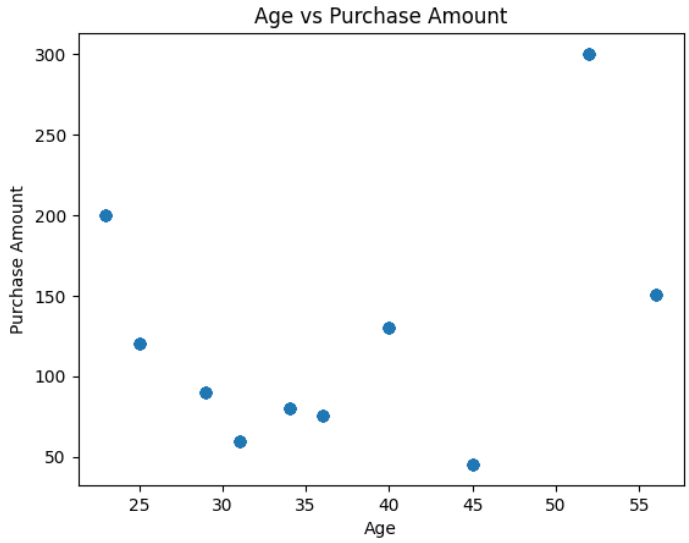

Figure 8. Extra Scatter Plot

6. Proposal: What models would you like to run on the data in Module C?

For the capstone project in Module C, more advanced machine learning models such as Gradient Boosting will be explored to improve prediction performance. This choice is based on the Random Forest results, which showed that certain features, particularly product category, strongly influence predictions.

Model performance will be evaluated using RMSE and R^2, which measure prediction accuracy and explanatory power. These metrics are appropriate for assessing how well the model predicts purchase amount and explains variation in the data.

This project is relevant to industry because businesses use customer data to improve marketing strategies, product recommendations, and pricing decisions. Validation will be performed using train-test splits and cross-validation to ensure reliable results. A successful model will show lower prediction error and consistent performance across different data samples.

In addition to Gradient Boosting, other models such as regression trees or ensemble methods could also be explored to further improve prediction performance. These models can help capture interactions between variables and provide more accurate predictions. It is also important to consider how the results will be used in practice. For example, businesses may prioritize models that are not only accurate but also interpretable. Therefore, balancing model performance with explainability will be an important consideration in the capstone project.

7. Teamwork: summarize how your team functioned as a group (e.g. giving feedback)

Throughout the semester, the team maintained consistent communication and completed all required tasks on time. Team members participated in weekly discussions and peer feedback activities, particularly through reviewing each other's homework and providing constructive comments.

The team followed the agreed expectations, and communication was timely when coordinating submissions and sharing work. There were no major issues with participation or deadlines, and all members contributed as expected.

Based on overall participation, communication, and contribution to group activities, all team members, including myself, would be assigned full points.

8. Weekly Graphs and Homework assignments

The weekly Jupyter Notebook assignments completed during Weeks 8 through 12 are included with this Final Project submission as separate files. These notebooks contain the code, visualizations, and multivariate and modeling analysis performed throughout the second half of the course.

The assignments include weekly graph questions, multivariate analysis techniques such as heatmaps, bubble plots, principal component analysis, and linear regression, as well as Random Forest modeling and storytelling visualizations. These notebook files provide detailed steps and supporting figures used in this report, while this summary document highlights the key insights and conclusions derived from that work.

9. AI appendix

AI tools were used to assist with structuring and refining the written content of this report. These tools helped improve clarity, organization, and wording.

AI was also occasionally used during the Jupyter Notebook assignments for minor coding support, including debugging, clarifying Python syntax, and generating visualization ideas. All analysis, modeling, and interpretation of results were completed independently.

10. References

Gondaliya, S. (2026). Customer purchase dataset [Data set]. Kaggle. https://www.kaggle.com/datasets/sapnagondaliya/customer-purchase/data

Knaflic, C. N. (2015). Storytelling with data: A data visualization guide for business professionals. John Wiley & Sons.

Pandas Development Team. (2023). pandas documentation. https://pandas.pydata.org/

Hunter, J. D. (2007). Matplotlib: A 2D graphics environment. Computing in Science & Engineering, 9(3), 90–95.

Waskom, M. (2021). Seaborn: Statistical data visualization. Journal of Open Source Software, 6(60), 3021.

Van Rossum, G., & Drake, F. L. (2009). Python 3 reference manual. CreateSpace.